Car2 / CAN network and controller wiring

Physical cable order supplied by user: acceleration -> master -> steering -> brake

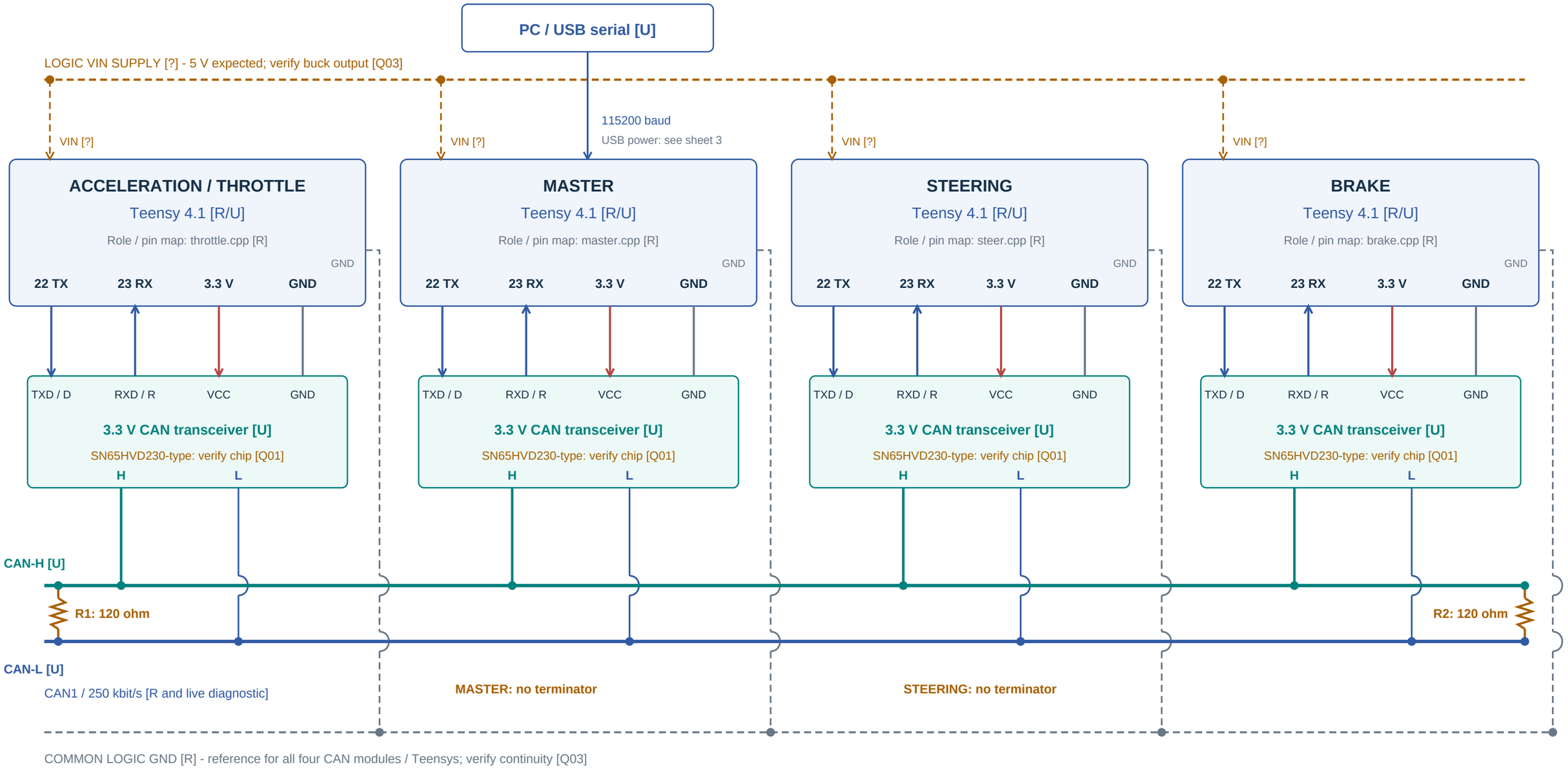
DRAFT r01 / REVIEW

[U] user report or live observation

[R] repository / intended wiring

[?] needs confirmation

Dashed amber = unverified detail



Q02 / Termination needs physical verification

Two 120-ohm resistors only: acceleration end and brake end. Count module jumpers. Expect about 60 ohms H-L with ALL power and USB off. Earlier reported measurement: about 30 ohms.

Brake-end condition / reported disconnected Teensy

Brake CAN module is still on H/L [U], so it remains the cable end. Its terminator still counts with the Teensy unplugged. If the entire brake cable is removed, steering becomes the new terminated end.

Basis: calebhylkema/car2_ws pin map + current session reports/tests. Reference repo does not override these pins.

Review draft: [?] details are incomplete. Diagram colors do not specify vehicle wire colors. Junction dots indicate connections.

Car2 / Actuator control and feedback wiring

Pin numbers follow the Caleb repository. They describe intended wiring, not a completed continuity audit.

DRAFT r01 / REVIEW

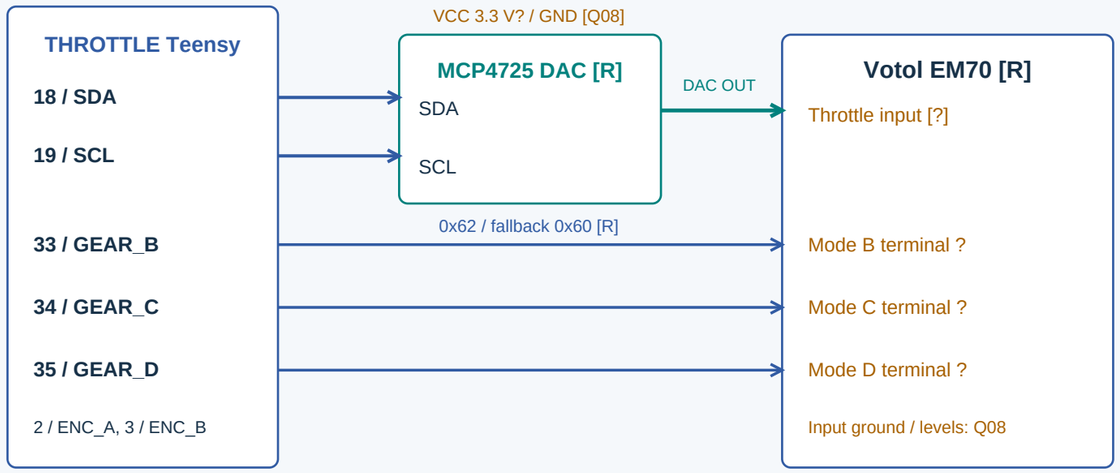
[U] user report or live observation

[R] repository / intended wiring

[?] needs confirmation

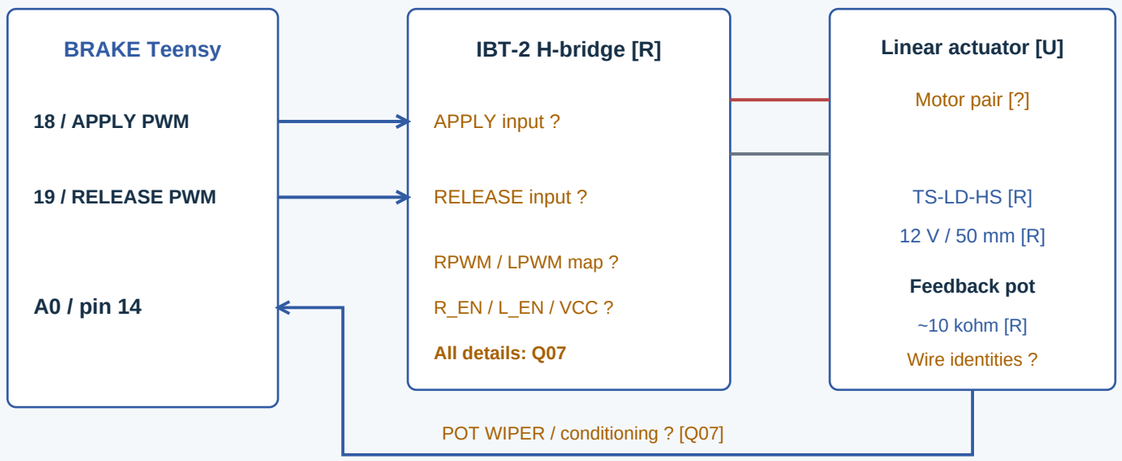
Dashed amber = unverified detail

ACCELERATION / THROTTLE [R]



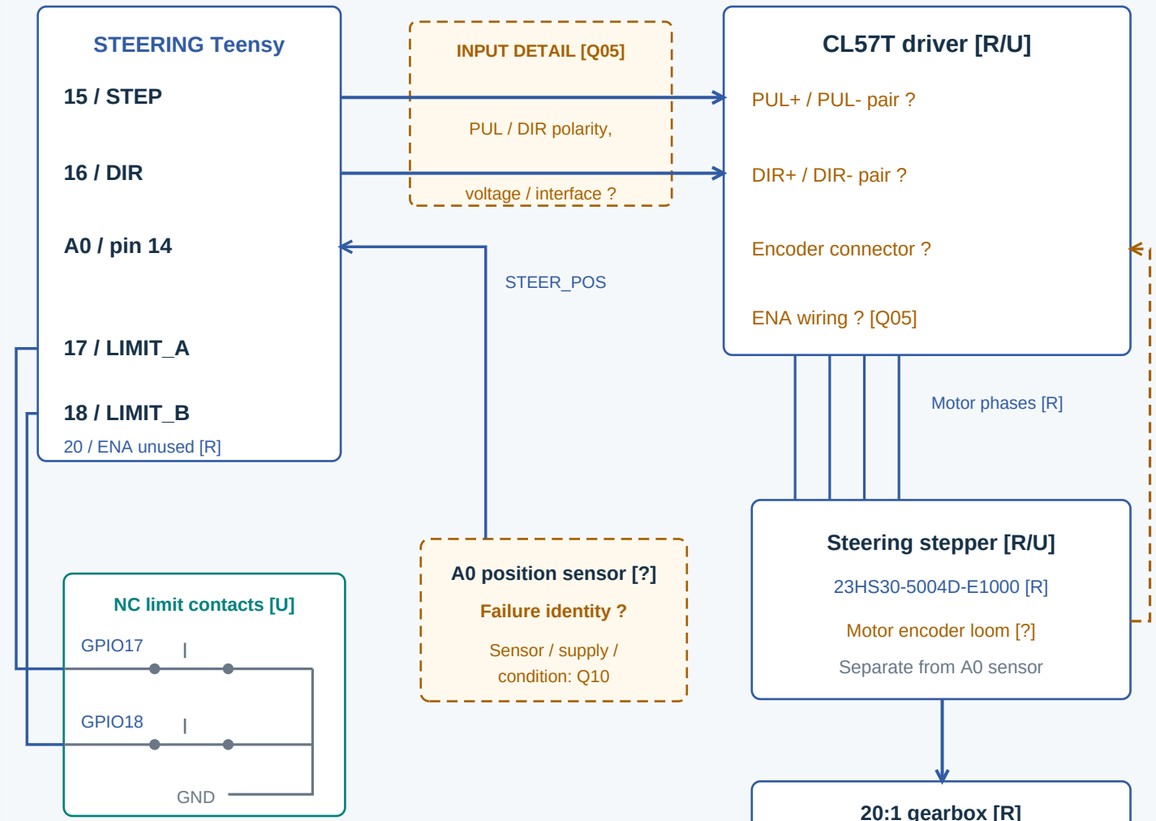
[R] DAC idle 0.80 V; maximum 3.30 V. LPD3806 encoder: A -> pin 2, B -> pin 3; encoder supply/conditioning needs confirmation [Q10].

BRAKE / LINEAR ACTUATOR [R/U]



[R] Position feedback decreases while applying. Firmware ADC targets: 945 released, 755 applied (745 mechanical limit). Pot supply and pin scaling require verification.

STEERING [R/U]



[Q06] Which contact is reached in each steering travel direction is still unverified. See observed pin/location mismatch below.

Q06 / Steering switch observations - do not assume side = travel

GPIO17: firmware calls LEFT; manually holding the right-side switch activated this pin [U].
GPIO18: firmware calls RIGHT; manually holding the left-side switch activated this pin [U].
Released = LOW; pressed = HIGH [U]. Actual mechanical travel direction must be checked before using either contact as a directional stop.
Q10: a failed steering encoder is reported; A0 sensor vs motor-encoder identity is unconfirmed.

Basis: calebhylkema/car2_ws pin map + current session reports/tests. Reference repo does not override these pins.

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Car2 / Power distribution and review checklist

Power topology is incomplete. Solid reported paths and dashed unverified paths are intentionally distinguished.

DRAFT r01 / REVIEW

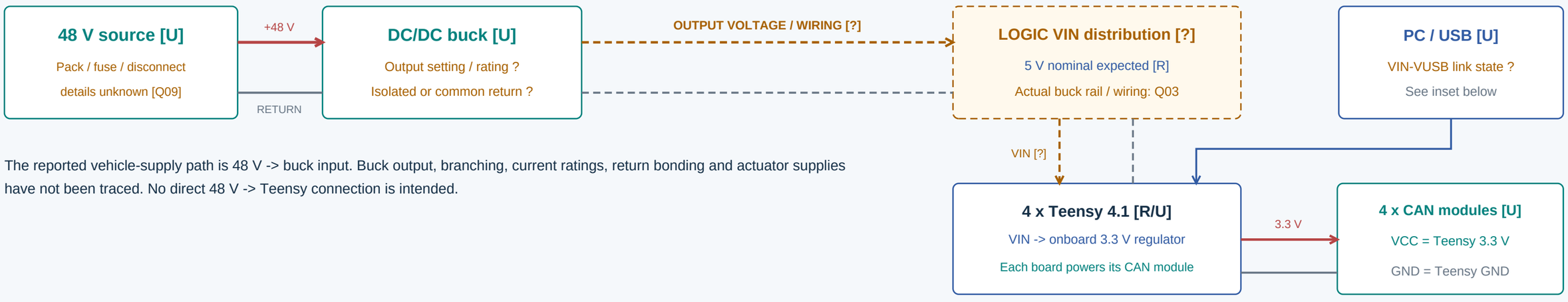
[U] user report or live observation

[R] repository / intended wiring

[?] needs confirmation

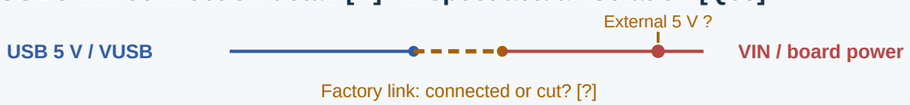
Dashed amber = unverified detail

LOGIC POWER / SOURCE AND DISTRIBUTION TO VERIFY [Q03]



The reported vehicle-supply path is 48 V -> buck input. Buck output, branching, current ratings, return bonding and actuator supplies have not been traced. No direct 48 V -> Teensy connection is intended.

USB / VIN connection detail [R] - inspect actual isolation [Q03]



PJRC: VIN and VUSB are normally linked. USB plus external VIN requires appropriate isolation of that power connection. Inspect the installed boards; keep USB data and signal ground correctly connected.

ACTUATOR SUPPLIES / NO UNVERIFIED POWER JUMPERS DRAWN

Votol EM70: traction supply, fuse, enable and motor phases [?].

CL57T: actual supply voltage, polarity and driver version [?].

IBT-2 brake: repository specifies a fused 12 V motor supply [R]; source and fuse rating [?]. Logic VCC and enable wiring [?].

CORRECTIONS TO REQUEST - reply using Q01 ... Q10

Q01	CAN module chip/model; actual H/L screw-terminal labels and wire polarity.
Q02	Which termination jumpers/resistors are fitted? Confirm 60 ohms powered off.
Q03	Buck output(s), VIN wiring, ground bonding and each VIN-VUSB link state.
Q04	Driver/controller model labels and all motor-supply voltages/fuse ratings.
Q05	CL57T PUL+/PUL-, DIR+/DIR-, ENA connections and any interface circuit.
Q06	Which pin17/pin18 switch is physically reached on left and right travel?
Q07	IBT-2 PWM/enable/VCC terminals; actuator motor pair and feedback wiring.
Q08	MCP4725 supply; Votol throttle ground/input and GPIO33/34/35 terminals.
Q09	Physical E-stop/disconnect/contactor chain; underglow supply and switching.
Q10	Which steering sensor failed? Motor encoder vs A0 feedback; encoder supplies.

Basis: calebhylkema/car2_ws pin map + current session reports/tests. Reference repo does not override these pins.

Review draft: [?] details are incomplete. Diagram colors do not specify vehicle wire colors. Junction dots indicate connections.